

Error Detection and Correction Techniques

BLM3051
Data
Communication
Week 5

- Data Link Layer (in OSI model)
- Error reasons
 - Attenuation
 - Delay Distortion
 - Video + Voice
 - Problem in time sensitive conditions
 - Noise in the communication environment
 - Thermal noise
 - Random electron motion
 - Intermodulation noise
 - CrossTalk
 - Impulse Noise

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Error Types

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- Single bit error
- Multi bit error
- Error bursts

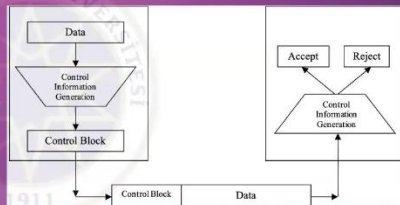
Data Sent	Data Received
1 0 0 0 0 1 1 1	1 0 1 0 0 1 1 1
1 0 0 0 0 1 1 1	1 0 1 0 0 0 1 1
1 0 0 0 0 1 1 1	1 0 0 1 1 0 0 1

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Error Detection

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- Both sides have original data?
- Sending data twice?
- Control block?
 - 4 different types
 - VRC (Vertical Redundancy Code)
 - LRC (Longitudinal Redundancy Code)
 - CRC (Cyclic Redundancy Check)
 - Checksum



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VRC (Vertical Redundancy Code)

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- Parity check
- Simple error coding technique
- The number of errors should be odd.
- XOR operation

Data Sent	VRC	Data
1 0 1 0 0 1 1 0	1	0 1 0 0 1 1 0
Data Received 1	VRC	Data
1 0 1 0 0 1 1 0	1	0 1 0 0 1 1 0
Data Received 2	VRC	Data
1 0 1 1 0 1 1 0	1	0 1 0 1 1 1 0
Data Received 3	VRC	Data
1 0 1 1 0 1 0 0	1	0 1 0 1 0 0 0

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LRC (Longitudinal Redundancy Code)

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- LRC is 2D-VRC

	Byte 1	Byte 2	Byte 3	Byte 4	LRC
	1	0	1	1	1
	0	0	1	1	0
	0	1	0	1	0
	1	1	0	1	1
	1	1	1	0	0
	0	1	1	0	0
	1	0	0	0	1
VRC	0	1	0	0	1

10011010	00110101	11001100	11110000	10010011
10011010	01110111	11001100	10110010	10010011

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CRC (Cyclic Redundancy Check)

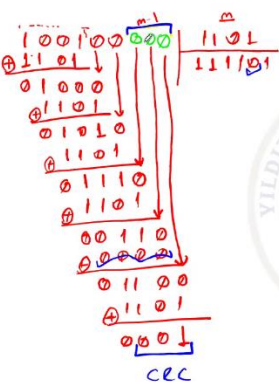
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- The data to be sent is divided into a predetermined prime polynomial.
- The remainder value is added to the data to be sent as an error control code.
- The remainder zero in receiver side means that error-free transmission.
- Common polynomials used for CRC: 13-bits, 17-bits, 33-bits
 - The number of undetectable errors is almost zero
- Commonly used polynomials in CRC technique:
 - CRC-12 $x^{12} + x^{11} + x^3 + x + 1$
 - CRC-16 $x^{16} + x^{15} + x^2 + 1$
 - CRC-ITU $x^{16} + x^{12} + x^5 + 1$
 - CRC-32 $x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$

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Example: Data Sent: 100100, polynom: $x^3 + x^2 + 1$, CRC = ?



Checksum

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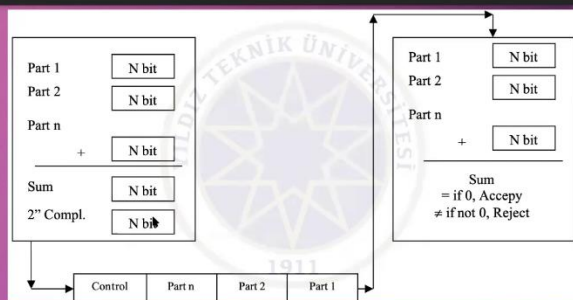
- The sender divides the data into N-bits parts (usually 16 bits are used).
- The parts are collected using the first complementary arithmetic.
 - In this way, a total value of only N bits is obtained.
- Calculate two's complement using summed value
 - The calculated value is added to the end of the information to be sent.
- The checksum detects all of the odd errors and most of the even numbers.
 - However, if one or more bits in a part are 0 when they are 1, but there is a 0 when 1 in another part, the error will not be understood because there will be no difference in this column sum.

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Checksum - Con't

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Hamming Code

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- If we sent m bit data, the error occurs in $1, 2, \dots, m$ bit
- Adding error-free state, the data length will be $m+1$
- Control block length must be $\log_2(m+1) \leq r$
- $m+r$ bit must be sent error-free
- So, control block length must be $\log_2(m+r+1) \leq r$
- (1, 2, 4, 8, 16, bits)

B ₁₁	B ₁₀	B ₉	B ₈	B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁
D ₇	D ₆	D ₅	R ₄	D ₄	D ₃	D ₂	R ₃	D ₁	R ₂	R ₁

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Hamming Code

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- $m+r$ bit must be sent error-free
- So, control block length must be $\log_2(m+r+1) \leq r$
- (1, 2, 4, 8, 16, bits)

	R ₄	R ₃	R ₂	R ₁	Info
0	0	0	0	0	Error-free
1	0	0	0	1	1. bit error
2	0	0	1	0	2. bit error
3	0	0	1	1	3. bit error
4	0	1	0	0	4. bit error
5	0	1	0	1	5. bit error
6	0	1	1	0	6. bit error
7	0	1	1	1	7. bit error
8	1	0	0	0	8. bit error
9	1	0	0	1	9. bit error
10	1	0	1	0	10. bit error
11	1	0	1	1	11. bit error

B ₁₁	B ₁₀	B ₉	B ₈	B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁
D ₇	D ₆	D ₅	R ₄	D ₄	D ₃	D ₂	R ₃	D ₁	R ₂	R ₁

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Hamming Code - Con't

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- $R_1 = B_1 \oplus B_3 \oplus B_5 \oplus B_7 \oplus B_9 \oplus B_{11}$
- $R_2 = B_2 \oplus B_3 \oplus B_6 \oplus B_7 \oplus B_{10} \oplus B_{11}$
- $R_3 = B_4 \oplus B_5 \oplus B_6 \oplus B_7$
- $R_4 = B_8 \oplus B_9 \oplus B_{10} \oplus B_{11}$

	R ₄	R ₃	R ₂	R ₁	Info
0	0	0	0	0	Error-free
1	0	0	0	1	1. bit error
2	0	0	1	0	2. bit error
3	0	0	1	1	3. bit error
4	0	1	0	0	4. bit error
5	0	1	0	1	5. bit error
6	0	1	1	0	6. bit error
7	0	1	1	1	7. bit error
8	1	0	0	0	8. bit error
9	1	0	0	1	9. bit error
10	1	0	1	0	10. bit error
11	1	0	1	1	11. bit error

B ₁₁	B ₁₀	B ₉	B ₈	B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁
D ₇	D ₆	D ₅	R ₄	D ₄	D ₃	D ₂	R ₃	D ₁	R ₂	R ₁

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Hamming Code - Con't

- $R_1 = B_1 \oplus B_3 \oplus B_5 \oplus B_7 \oplus B_9 \oplus B_{11}$
- $R_2 = B_2 \oplus B_3 \oplus B_6 \oplus B_7 \oplus B_{10} \oplus B_{11}$
- $R_3 = B_4 \oplus B_5 \oplus B_6 \oplus B_7$
- $R_4 = B_8 \oplus B_9 \oplus B_{10} \oplus B_{11}$

B ₁₁	B ₁₀	B ₉	B ₈	B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁
1	0	0		1	1	0		1		

- $R_1 = B_3 \oplus B_5 \oplus B_7 \oplus B_9 \oplus B_{11} = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 1$
- $R_2 = B_3 \oplus B_6 \oplus B_7 \oplus B_{10} \oplus B_{11} = 1 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 0$
- $R_3 = B_5 \oplus B_6 \oplus B_7 = 0 \oplus 1 \oplus 1 = 0$
- $R_4 = B_9 \oplus B_{10} \oplus B_{11} = 0 \oplus 0 \oplus 1 = 1$

	R ₄	R ₃	R ₂	R ₁	Info
0	0	0	0	0	Error-free
1	0	0	0	1	1. bit error
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4	0	1	0	0	4. bit error
5	0	1	0	1	5. bit error
6	0	1	1	0	6. bit error
7	0	1	1	1	7. bit error
8	1	0	0	0	8. bit error
9	1	0	0	1	9. bit error
10	1	0	1	0	10. bit error
11	1	0	1	1	11. bit error

Hamming Code

- If we sent m bit data, the error occurs in 1,2,...,m bit